

Database Management Systems

UE24CS351A - Functional Dependencies

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PES UNIVERSITY

Database Management Systems

Unit 3: Design Concepts and Implementation

Slides adapted from Author Slides of "Database System Concepts", Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems", Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

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Functional Dependencies

- Each relation schema consists of a number of attributes, and the relational database schema consists of a number of relation schemas.
- We have assumed that attributes are grouped to form a relation schema by using the common sense of the database designer or by mapping a database schema design from a conceptual data model such as the ER data model.
- We did not develop **any measure of appropriateness or goodness to measure the quality of the design**, other than the intuition of the designer.
- There are **two levels at which we can discuss the goodness of relation schemas**.
  - The first is the **logical (or conceptual) level**—how users interpret the relation schemas and the meaning of their attributes.
    - The second is the **implementation (or physical storage) level**—how the tuples in a base relation are stored and updated.

Functional Dependencies

- The theory is applicable primarily to the top-down design approach, and as such is more appropriate when performing design of databases by analysis and decomposition of sets of attributes that appear together in files, in reports, and on forms in real-life situations.
- Relational database design ultimately produces a set of relations.
  - The implicit goals of the design activity are information preservation and minimum redundancy.
- **A functional dependency is a constraint between two sets of attributes from the database.**
- Suppose that our relational database schema has n attributes A1, A2, … , An; let us think of the whole database as being described by a single universal relation schema R = {A1, A2, … , An}.
- We do not imply that we will actually store the database as a single universal table;
- We use this concept only in developing the formal theory of data dependencies

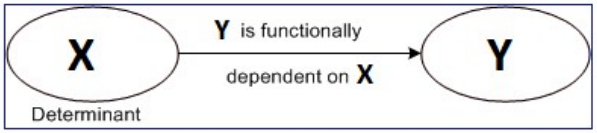
Functional Dependencies

- **Functional dependencies (FDs)**: A Functional Dependency is a constraint between two sets of attributes from the database.

They are:

  - Used to specify *formal measures* of the "goodness" of relational designs
  - Are used to define **normal forms** for relations
  - Are **constraints** that are derived from the *meaning* and *interrelationships* of the data attributes

- The functional dependency (FD) is a relationship that exists between two attributes, Where one attribute can directly or indirectly derived from the other attribute.
- **A set of attributes X *functionally determines* a set of attributes Y if the value of X determines a unique value for Y**



Functional dependency between X and Y

Dependent

Defining Functional Dependencies

- A functional dependency, denoted by X → Y, between two sets of attributes X and Y that are subsets of R specifies a constraint on the possible tuples that can form a relation state r of R.
  - The constraint is that, for any two tuples t1 and t2 in r that have t1[X] = t2[X], they must also have t1[Y] = t2[Y].

- *X functionally determines Y in a relation schema R if, and only if, whenever two tuples of r(R) agree on their X-value, they must necessarily agree on their Y-value.*

Note the following:

- If a constraint on R states that there cannot be more than one tuple with a given X-value in any relation instance r(R)—that is, **X is a candidate key of R**—this implies that X → Y for any subset of attributes Y of R (because the key constraint implies that no two tuples in any legal state r(R) will have the same value of X).
- If X is a candidate key of R, then X → R.
- If X → Y in R, this does not say whether or not Y → X in R

Use of Functional Dependencies

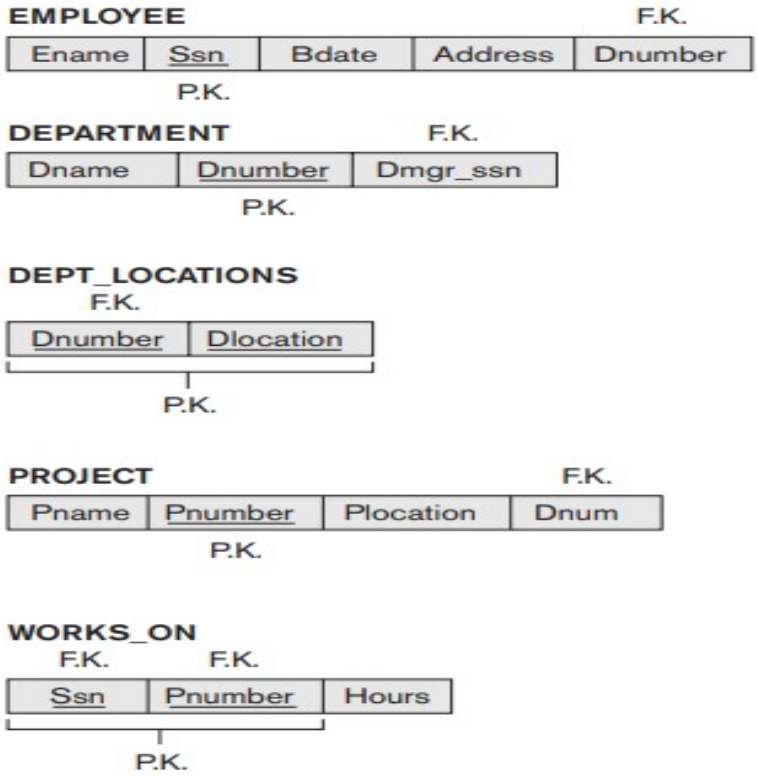
To Summarize

- X → Y in R specifies a *constraint* on all relation instances r(R)
- Written as X → Y; can be displayed graphically on a relation schema
- FDs are derived from the real-world constraints on the attributes
- A functional dependency is a **property of the semantics or meaning of the attributes.**
- **Relation extensions r(R)** that satisfy the functional dependency constraints are called

  legal relation states (or legal extensions) of R.

Examples of Functional dependency

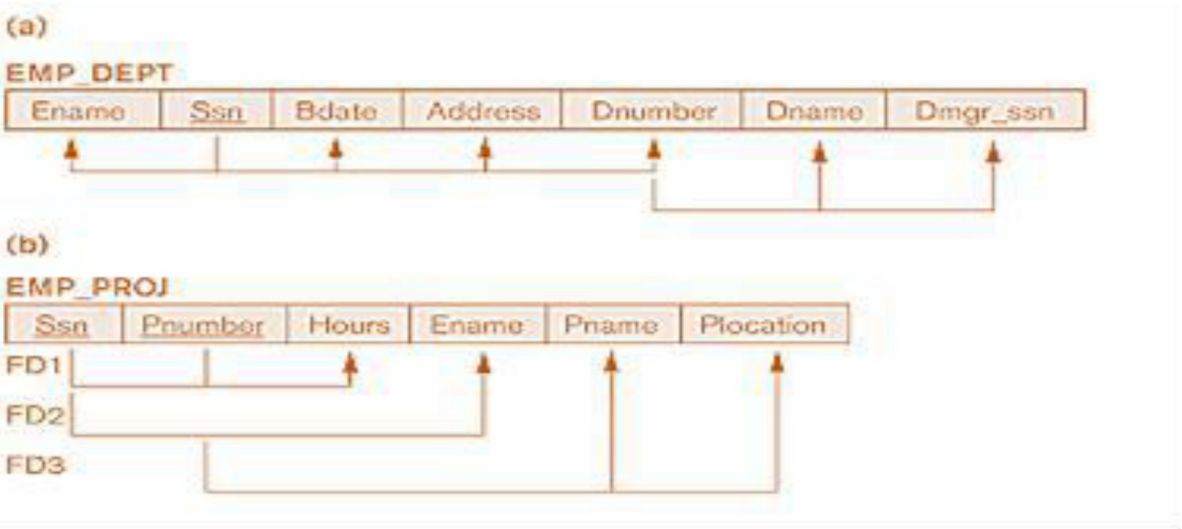
- Social security number determines employee name
  - SSN → ENAME
- Project number determines project name and location
  - PNUMBER → {PNAME, PLOCATION}
- Employee ssn and project number determines the hours per week that the employee works on the project
  - {SSN, PNUMBER} → HOURS



Diagrammatic Notation of FDs

- Each FD is displayed as a horizontal line.
- The left hand side attributes are connected by vertical lines to the line representing the FD, whereas the right hand side attributes are connected by the lines with arrows pointing toward the attributes.

Try and take a guess at what these FDs are?



Defining Functional Dependencies

Employee_ID	Employee_Name	Department	Salary
101	John	HR	50000
102	Mary	Finance	60000
103	Sarah	HR	55000

- In this example, let's say we want to explore the functional dependency
- "Department → Salary," which means that for any two tuples with the same value for the "Department" attribute, they must have the same value for the "Salary" attribute.

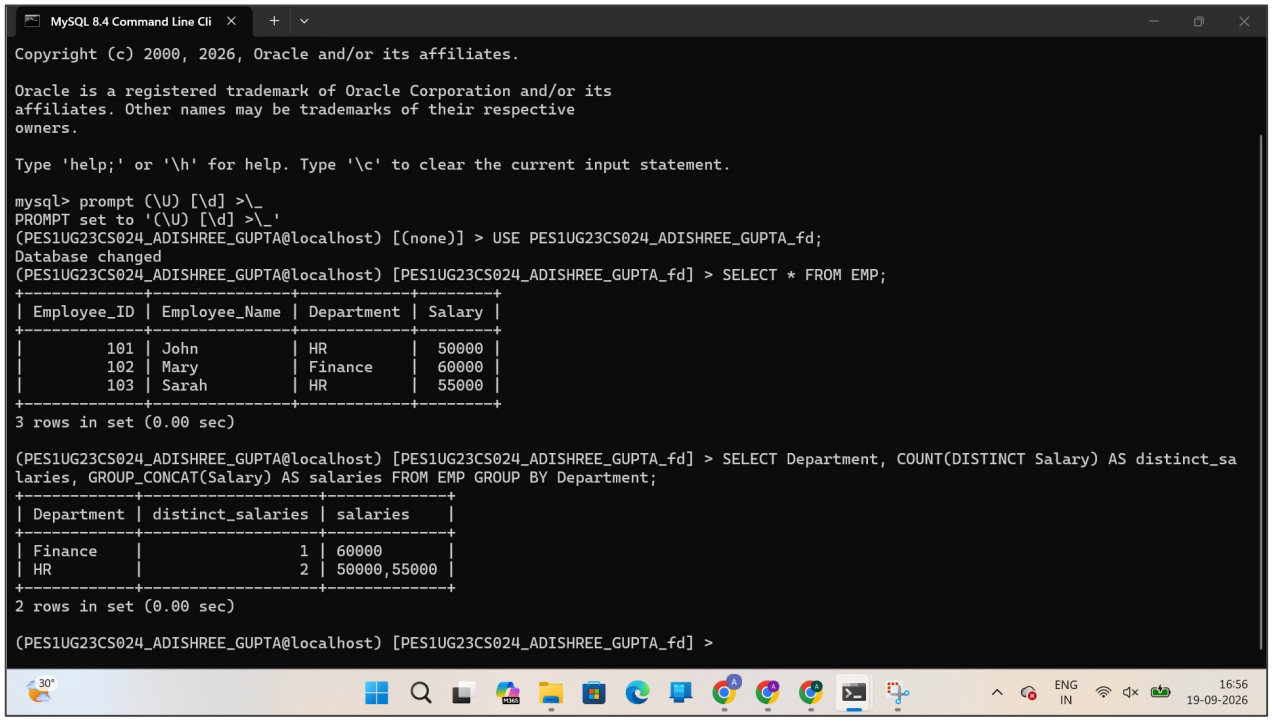
Is the functional dependency satisfied?

MySQL Execution Verification

Slide 10 - answer to 'Is the functional dependency satisfied?'

SQL executed in the MySQL client

```
USE PES1UG23CS024_ADISHREE_GUPTA_fd;
SELECT * FROM EMP;
SELECT Department,
        COUNT(DISTINCT Salary) AS distinct_salaries,
        GROUP_CONCAT(Salary) AS salaries
FROM EMP
GROUP BY Department;
```



```
MySQL 8.4 Command Line Cli

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> prompt (\U) [\d] >\_
PROMPT set to '(\U) [\d] >\_'
(PES1UG23CS024_ADISHREE_GUPTA@localhost) [(none)] > USE PES1UG23CS024_ADISHREE_GUPTA_fd;
Database changed
(PES1UG23CS024_ADISHREE_GUPTA@localhost) [PES1UG23CS024_ADISHREE_GUPTA_fd] > SELECT * FROM EMP;
+-------------+---------------+------------+--------+
| Employee_ID | Employee_Name | Department | Salary |
+-------------+---------------+------------+--------+
|         101 | John          | HR         |  50000 |
|         102 | Mary          | Finance    |  60000 |
|         103 | Sarah         | HR         |  55000 |
+-------------+---------------+------------+--------+
3 rows in set (0.00 sec)

(PES1UG23CS024_ADISHREE_GUPTA@localhost) [PES1UG23CS024_ADISHREE_GUPTA_fd] > SELECT Department, COUNT(DISTINCT Salary) AS distinct_sa
laries, GROUP_CONCAT(Salary) AS salaries FROM EMP GROUP BY Department;
+------------+------------------+-------------+
| Department | distinct_salaries | salaries    |
+------------+------------------+-------------+
| Finance    |                1 | 60000       |
| HR         |                2 | 50000,55000 |
+------------+------------------+-------------+
2 rows in set (0.00 sec)

(PES1UG23CS024_ADISHREE_GUPTA@localhost) [PES1UG23CS024_ADISHREE_GUPTA_fd] >
```

Department HR carries two different salaries, 50000 and 55000, so distinct_salaries = 2. Two tuples agree on Department but disagree on Salary, which is exactly the condition that breaks an FD. Department -> Salary therefore does NOT hold - the answer to the question posed on slide 10.

Examples of FD constraints (2)

- **The main use of FD is to describe further a relational schema R by specifying constraints on its attributes that must hold all the times.**

- **The constraint must hold on *every* relation instance r(R)**

- If K is a key of R, then **K functionally determines all attributes in R**
  - (since we never have two distinct tuples with t1[K]=t2[K])

Terms

- An attribute of relation schema R is called a **prime attribute** of R if it is a member of some candidate key of R.
- An attribute is called **non prime** if it is not a prime attribute—that is, if it is not a member of any candidate key.
- Both Ssn and Pnumber are prime attributes of WORKS_ON, whereas other attributes of WORKS_ON are nonprime

Types of Functional Dependency

Full Functional Dependency:

- A functional dependency X → Y is said to be a full functional dependency if removing any attribute A from X causes the dependency to no longer hold.
  - In other words, for every A ∈ X, (X − {A}) does not functionally determine Y.

Example:

- Consider the dependency ABC → D, meaning ABC determines D.
- None of the proper subsets BC → D, C → D, or A → D can determine D.
- Therefore, D is fully functionally dependent on ABC.

Another example:

- {Ssn, Pnumber} → Hours is a full functional dependency because neither Ssn → Hours nor Pnumber → Hours holds true individually.

Types of Functional Dependency

Partial Functional Dependency:

- A functional dependency X → Y is said to be a partial dependency if some attribute A ∈ X can be removed from X and the dependency still holds; in other words, for some A ∈ X, (X − {A}) → Y remains valid.

- In simpler terms, if a non-prime attribute (an attribute that is not part of any candidate key) is functionally dependent on only a portion of the candidate key, rather than the entire key, the dependency is referred to as a Partial Dependency.

- **Example:** Consider the following
- {Ssn, Pnumber} → Ename is partial, because Ssn → Ename holds.

Types of Functional Dependency

Transitive Functional Dependency:

- A transitive functional dependency occurs in a relation when a non-prime attribute (an attribute that is not part of any candidate key) depends on another non-prime attribute, rather than depending directly on the primary key.

- In other words, if a non-prime attribute is derived indirectly through another attribute (which itself depends on a key), the dependency is transitive.

- Formally, if X → Y and Y → Z, then X → Z is a transitive dependency, provided that X is a key and Z is a non-prime attribute.

Types of Functional Dependency

Trivial Functional Dependency:

- A trivial functional dependency is one that holds by definition, based on the Reflexive Rule of functional dependencies.
- It states that if Y is a subset of X, then the dependency X → Y is always true (trivial).

Example:

- If X = {A, B, C} and Y = {B, C}, then ABC → BC is a trivial functional dependency, because Y (BC) is a subset of X (ABC).

Types of Functional Dependency

Non Trivial Functional Dependency:

- A non-trivial functional dependency is one in which the dependent attribute (Y) is not a subset of the determinant (X).
- In other words, the dependency X → Y is non-trivial if and only if

  $Y \not\subseteq X$ ie Y is not a subset of X

- This means that Y contains at least one attribute that is not already part of X — so the dependency provides new information about the relationship between attributes.

Ruling Out FDs

- Consider the following TEACH relation.

we can say that the FD: Text → Course may exist.

- However, the FDs Teacher → Course, Teacher → Text, Course → Text are ruled out.
- (This is because the element on the LHS have similar values in the current column, but are not the same on the RHS column)

TEACH

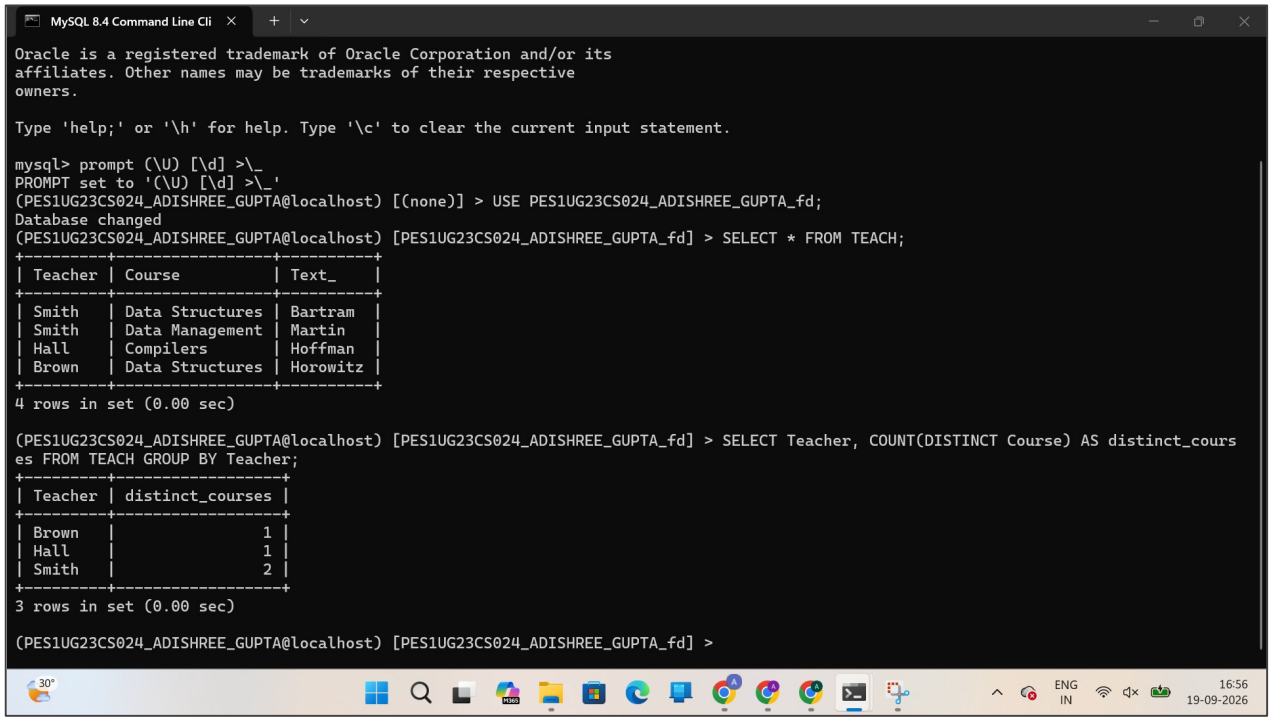
Teacher	Course	Text
Smith	Data Structures	Bartram
Smith	Data Management	Martin
Hall	Compilers	Hoffman
Brown	Data Structures	Horowitz

MySQL Execution Verification

Slide 18 - ruling out FDs in the TEACH relation

SQL executed in the MySQL client

```
USE PES1UG23CS024_ADISHREE_GUPTA_fd;
SELECT * FROM TEACH;
SELECT Teacher,
        COUNT(DISTINCT Course) AS distinct_courses
FROM TEACH
GROUP BY Teacher;
```



Smith appears with two distinct courses, so Teacher -> Course is ruled out exactly as the slide states. Any teacher row showing a count above 1 is a live counterexample; Hall and Brown show 1 each, which is consistent but proves nothing on its own.

What FDs may exist?

- A relation *R*(A, B, C, D) with its extension.

- The following FDs may hold because the 4 tuples in the current extension have no violation of these constraints: B -> C; C->B; {A,B} -> C; {A,B} -> D; and {C,D} -> B
- The following do not hold: A-> B, B-> A ; D -> C

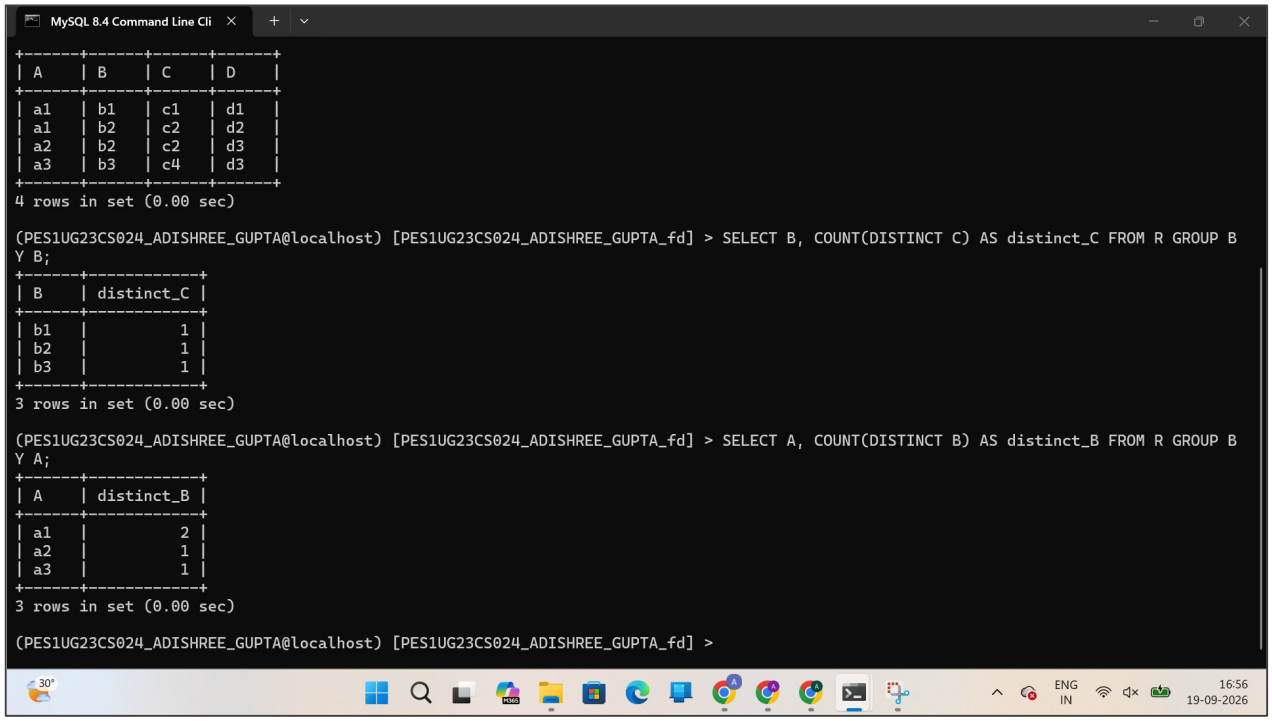
A	B	C	D
a1	b1	c1	d1
a1	b2	c2	d2
a2	b2	c2	d3
a3	b3	c4	d3

MySQL Execution Verification

Slide 19 - every FD claimed for R(A, B, C, D) tested on the real extension

SQL executed in the MySQL client

```
USE PES1UG23CS024_ADISHREE_GUPTA_fd;
SELECT * FROM R;
SELECT B, COUNT(DISTINCT C) AS distinct_C
FROM R GROUP BY B;
SELECT A, COUNT(DISTINCT B) AS distinct_B
FROM R GROUP BY A;
```



Grouping by B gives distinct_C = 1 for every value, so B -> C holds on this extension. Grouping by A gives distinct_B = 2 for a1, so A -> B is violated. These are two of the eight claims made on slide 19, tested directly against the four tuples.

Practice Questions - Functional Dependencies (1/3)

1. [MCQ] In the TEACH relation on slide 18 a student claims Course -> Teacher holds.

- **Which pair of tuples refutes the claim?**

(a) (Smith, Data Structures, Bartram) and (Smith, Data Management, Martin)
(b) (Smith, Data Structures, Bartram) and (Brown, Data Structures, Horowitz)
(c) (Hall, Compilers, Hoffman) and (Brown, Data Structures, Horowitz)
(d) No pair refutes it - the FD holds

- ***Answer:*** *(b). Both tuples agree on Course = 'Data Structures' but disagree on Teacher (Smith vs Brown), which is exactly the condition t1[X] = t2[X] and t1[Y] != t2[Y] that violates an FD. Option (a) shows two different courses, so it says nothing about Course -> Teacher.*

2. [Reasoning / comparison] WORKS_ON(Ssn, Pnumber, Hours, Ename) has candidate key {Ssn, Pnumber}.

- **Classify {Ssn, Pnumber} ->**

- **Hours and {Ssn, Pnumber} -> Ename as FULL or PARTIAL, and justify each.**
  - ***Answer:*** *{Ssn, Pnumber} -> Hours is FULL: removing either attribute breaks it, since neither Ssn -> Hours nor Pnumber -> Hours holds. {Ssn, Pnumber} -> Ename is PARTIAL: Ssn -> Ename already holds on its own, so Pnumber can be dropped.*
  - *Ename is a non-prime attribute determined by only part of the key.*

Practice Questions - Functional Dependencies (2/3)

3. [Critical thinking] Slide 19 says certain FDs 'may hold' on the basis of four tuples.

- **Explain why a relation instance can disprove an FD but can never prove one.**
- ***Answer:*** *A single pair of tuples that agree on X and disagree on Y is a counterexample, and one counterexample refutes the FD permanently.*
  - *But an FD is a constraint on EVERY legal state of R, while an instance shows only one state - a future tuple could still violate it. FDs come from the semantics of the attributes, not from the data currently stored.*

4. [MCQ] Which of the following is NOT a trivial functional dependency?

(a) {A, B, C} -> {B, C}
(b) {A, B} -> {A}
(c) {A, B} -> {C}
(d) {A} -> {A}

- ***Answer:*** *(c). By the reflexive rule an FD is trivial when the right-hand side is a subset of the left-hand side.*
  - *In (a), (b) and (d) it is. In (c) C is not among {A, B}, so the dependency is non-trivial and carries genuinely new information.*

Practice Questions - Functional Dependencies (3/3)

5. [Scenario / reasoning] In R(StudentID, Dept, DeptHead) with candidate key StudentID, both StudentID -> Dept and Dept -> DeptHead hold.

- **Is StudentID -> DeptHead a transitive dependency?**
- **State the condition that makes it one.**
- ***Answer:*** *Yes. StudentID -> Dept and Dept -> DeptHead give StudentID -> DeptHead.*
  - *It counts as TRANSITIVE because the intermediate attribute Dept and the determined attribute DeptHead are both non-prime - neither belongs to any candidate key - so DeptHead depends on the key only indirectly, through another non-prime attribute.*

6. [SQL / output prediction] Write one SQL query that decides whether B -> C holds in R(A, B, C, D), and state which result means the dependency holds.

- **What does the query return for the slide-19 data?**
- ***Answer:*** *SELECT B FROM R GROUP BY B HAVING COUNT(DISTINCT C) > 1; An EMPTY result means B -> C holds, because no B value is paired with more than one C value; any row returned is a counterexample.*
  - *On the slide-19 data it returns zero rows, confirming B -> C (verified on MySQL 8.4.6).*



THANK YOU

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